SHARP 2026 Projects in the Robotics and Motion Lab

Pranav A. Bhounsule
Associate Professor of Mechanical Engineering
University of Illinois at Chicago
Email: pranav@uic.edu
Lab Website: https://pab47.github.io

1 Effect of Backpack Load on Posture and Gait Stability During Walking

1.1 Project Overview

This project investigates how backpack weight affects posture and gait stability during walking using wearable sensors and motion analysis techniques.

1.2 Overarching Hypothesis

Increasing backpack weight increases forward trunk tilt and gait instability during walking.

1.3 Primary Research Methods and Techniques

Students will learn:

- Wearable biomechanics and gait analysis
- Human-subject experimental design
- Sensor calibration and motion tracking
- Time-series data collection and analysis

1.4 Tools and Technologies

Students will gain experience with:

- Arduino microcontrollers
- IMU sensors (accelerometer and gyroscope)
- Tilt sensors
- C programming in the Arduino environment
- Data logging and visualization using Python or Excel

1.5 Why This Hypothesis Matters

Understanding how external loads affect posture and walking mechanics can help improve ergonomic backpack design, reduce musculoskeletal strain, and contribute to wearable health-monitoring systems.

2 Biomechanical Analysis of Sit-to-Stand Motion Under Different Chair Heights

2.1 Project Overview

This project studies how chair height affects the biomechanics of sit-to-stand motion.

2.2 Overarching Hypothesis

Lower chair heights increase trunk acceleration and joint motion demands during sit-to-stand transitions.

2.3 Primary Research Methods and Techniques

Students will learn:

- Functional biomechanics
- Motion segmentation and event detection
- Joint-angle measurement
- Experimental repeatability analysis

2.4 Tools and Technologies

Students will gain experience with:

- Arduino microcontrollers
- IMU sensors
- Potentiometers for joint-angle measurement
- C programming
- Signal visualization and analysis tools

2.5 Why This Hypothesis Matters

Sit-to-stand motion is an important indicator of mobility and lower-body strength. This project contributes to rehabilitation engineering, assistive-device design, and ergonomic analysis.

3 Effect of Sensor-Assisted Cane Feedback on Walking Stability

3.1 Project Overview

This project evaluates whether sensor-assisted canes can improve walking stability during visually restricted walking tasks.

3.2 Overarching Hypothesis

Haptic feedback from a sensor-assisted cane reduces walking instability and obstacle collisions during visually restricted walking tasks.

3.3 Primary Research Methods and Techniques

Students will learn:

- Assistive robotics
- Human-in-the-loop experimentation
- Embedded sensing and feedback systems
- Navigation performance analysis

3.4 Tools and Technologies

Students will gain experience with:

- Arduino microcontrollers
- Ultrasonic sensors
- IMU sensors
- Haptic feedback systems using vibration motors
- C programming

3.5 Why This Hypothesis Matters

Assistive technologies can improve mobility and navigation safety. This project introduces students to rehabilitation robotics and accessibility-focused engineering.

4 Comparative Biomechanics of Stair Ascent, Stair Descent, and Level Walking

4.1 Project Overview

This project compares the biomechanics of stair ascent, stair descent, and level walking using wearable sensors.

4.2 Overarching Hypothesis

Stair descent produces larger impact accelerations and greater postural instability than stair ascent or level walking.

4.3 Primary Research Methods and Techniques

Students will learn:

- Comparative gait analysis
- Motion-event detection
- Acceleration and stability analysis
- Experimental comparison across locomotion modes

4.4 Tools and Technologies

Students will gain experience with:

- Arduino microcontrollers
- IMU sensors
- Tilt sensors
- C programming
- Signal analysis and plotting tools

4.5 Why This Hypothesis Matters

Understanding stair biomechanics is important for fall prevention, rehabilitation engineering, and wearable mobility-monitoring systems.

5 Effects of Walking Direction and Asymmetrical Loading on Gait Symmetry and Heart Rate

5.1 Project Overview

This project studies how backward walking and asymmetrical loading affect gait symmetry and exercise intensity.

5.2 Overarching Hypothesis

Backward walking and asymmetrical loading increase heart rate and gait asymmetry compared to normal forward walking.

5.3 Primary Research Methods and Techniques

Students will learn:

- Human gait analysis
- Exercise physiology
- Multi-sensor data collection
- Comparative motion studies

5.4 Tools and Technologies

Students will gain experience with:

- Arduino microcontrollers
- IMU sensors
- ECG or heart-rate sensors
- Weighted loading systems
- C programming
- Data analysis and visualization tools

5.5 Why This Hypothesis Matters

Backward walking is increasingly used in rehabilitation and sports training. This project investigates how altered walking conditions affect balance, coordination, and exercise intensity.

6 Effect of Smartphone Usage on Walking Posture and Stability

6.1 Project Overview

This project investigates how smartphone usage affects posture and gait stability during walking.

6.2 Overarching Hypothesis

Smartphone usage during walking increases forward trunk tilt and gait instability compared to normal walking.

6.3 Primary Research Methods and Techniques

Students will learn:

- Wearable sensing
- Human-factor biomechanics
- Dual-task motion analysis
- Experimental comparison methods

6.4 Tools and Technologies

Students will gain experience with:

- Arduino microcontrollers
- IMU sensors
- Tilt sensors
- C programming
- Motion-data visualization and analysis

6.5 Why This Hypothesis Matters

Distracted walking is associated with poor posture, increased instability, and pedestrian accidents. This project studies the biomechanical effects of smartphone usage in everyday life.

7 Comparative Postural Analysis of Standing and Sitting Desk Work

7.1 Project Overview

This project compares posture and stability during standing-desk and sitting-desk work conditions.

7.2 Overarching Hypothesis

Standing desks reduce forward trunk tilt but increase postural sway compared to seated desk work.

7.3 Primary Research Methods and Techniques

Students will learn:

- Ergonomic analysis
- Wearable posture sensing
- Long-duration human-subject monitoring
- Statistical comparison methods

7.4 Tools and Technologies

Students will gain experience with:

- Arduino microcontrollers
- IMU sensors
- Tilt sensors
- C programming
- Data analysis and visualization tools

7.5 Why This Hypothesis Matters

Poor workplace ergonomics contribute to chronic musculoskeletal issues. This project studies evidence-based approaches for improving workplace health and posture.

8 Effectiveness of Real-Time Biofeedback for Reducing Slouched Posture

8.1 Project Overview

This project develops and evaluates a wearable posture-monitoring device with real-time haptic feedback.

8.2 Overarching Hypothesis

Real-time haptic biofeedback reduces the duration and frequency of slouched posture during seated work.

8.3 Primary Research Methods and Techniques

Students will learn:

- Biofeedback systems
- Human-machine interaction
- Wearable rehabilitation engineering
- Real-time embedded control systems

8.4 Tools and Technologies

Students will gain experience with:

- Arduino microcontrollers
- IMU sensors
- Tilt sensors
- Vibration motors for haptic feedback
- C programming
- Motion-data analysis tools

8.5 Why This Hypothesis Matters

Wearable biofeedback systems can help improve posture habits, reduce musculoskeletal strain, and promote healthier workplace behaviors.